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Cloud Storage Technologies: Object, Block, and File Storage Compared

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CSA1511 :CLOUD COMPUTING AND BIG DATA ANALYTICS FOR
ENGINEERS

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Technical Presentation



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Introduction

What is Cloud Storage?

Cloud storage is a service that stores digital data on remote servers accessed through the Internet.

Why Cloud Storage?

- **Anywhere Access** – Access files securely from any device with an Internet connection.
- **Scalability** – Easily increase or decrease storage capacity based on demand.
- **Cost Efficiency** – Pay only for the storage resources you use.
- **Data Backup & Recovery** – Protects data through automatic backups and disaster recovery.
- **High Availability** – Ensures data is available with minimal downtime.
- **Data Security** – Provides encryption, authentication, and access control.



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Core Technical Explanation

Types of Cloud Storage

Object Storage

Stores data as objects

Each object has unique ID and metadata

Best for unstructured data

Block Storage

Stores data in fixed-size blocks

High performance

Used for databases and virtual machines

File Storage

Stores data as files and folders

Uses hierarchical directory structure

Shared across multiple users



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IoT-Cloud Architecture

Object Storage Architecture

Files



Objects



Metadata



Cloud Object Storage

Block Storage Architecture

Application



Operating System



Blocks



Physical Storage

File Storage Architecture

Users



Shared Folder



Directories

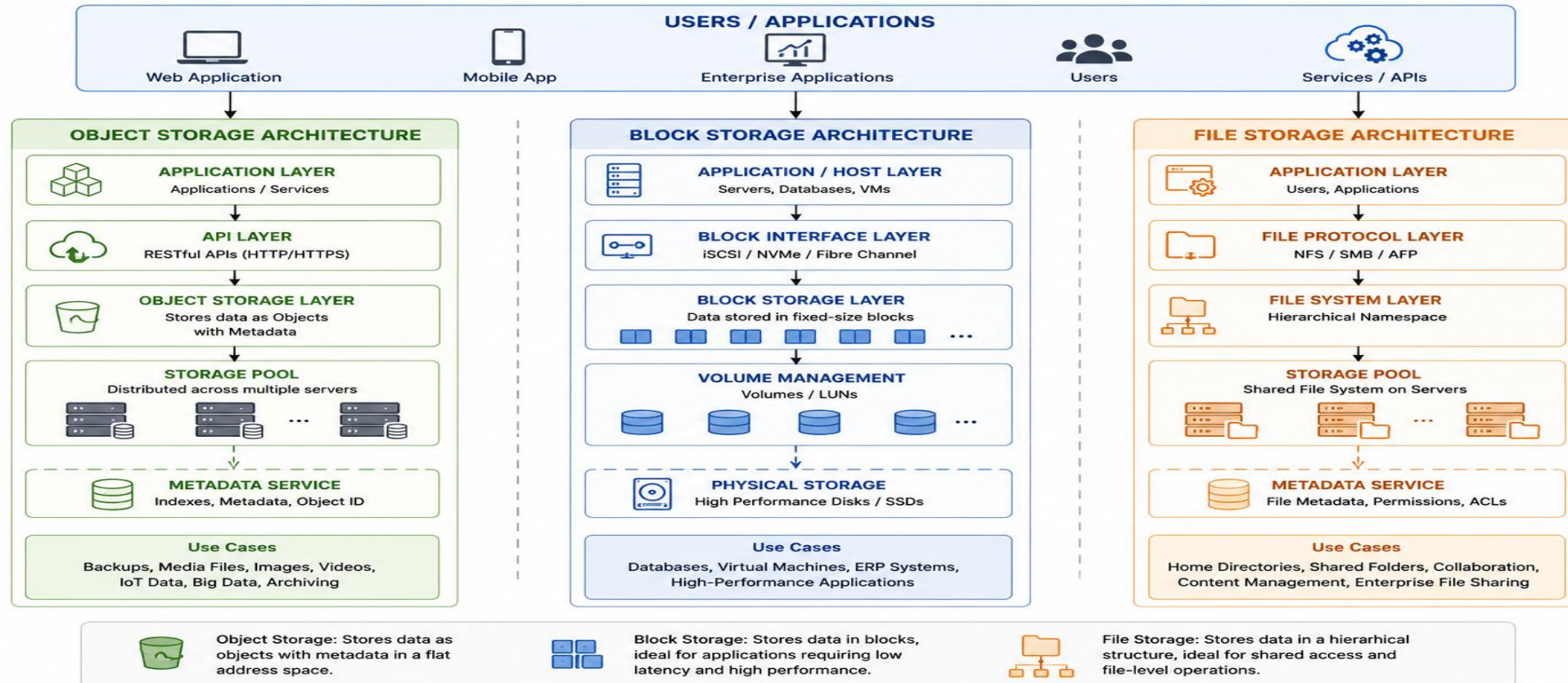


Files

Architecture Diagram

CLOUD STORAGE ARCHITECTURE

Object Storage vs Block Storage vs File Storage





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ADVANTAGES AND LIMITATIONS

Advantages

- High availability
- Scalable
- Secure
- Easy backup
- Global accessibility

Limitations

- Internet dependency
- Security concerns
- Vendor lock-in
- Data transfer cost



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Real-World Applications

Amazon AWS

Amazon S3 → Object Storage

Amazon EBS → Block Storage

Amazon EFS → File Storage

Microsoft Azure

Blob Storage

Managed Disks

Azure Files

Google Cloud

Cloud Storage

Persistent Disk

Filestore



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Example Use Case

Object Storage

Cloud backup
Images
Videos
IoT data
Big Data

Block Storage

Databases
Virtual Machines
Enterprise Applications
Boot Volumes

File Storage

Office Documents
Team Collaboration
Shared Drives
Home Directories



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Conclusion and Key Takeaways

Object Storage is best for large unstructured data.

Block Storage offers the highest performance.

File Storage is ideal for shared file access.

Choosing the right storage depends on application requirements, cost, and performance.

Key Takeaways

- ✓ Object → Scalable
- ✓ Block → Fast
- ✓ File → Easy Sharing



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Future Scope

AI-Driven Storage Management

Artificial Intelligence will automatically optimize data placement, predict storage requirements, detect anomalies, and improve overall storage efficiency.

2. Intelligent Data Tiering

Future cloud storage systems will automatically move frequently accessed data to high-speed storage and archive less-used data to lower-cost storage, reducing operational costs.

3. Edge Cloud Storage

Storage resources will be deployed closer to users and IoT devices, enabling faster data access, reduced latency, and real-time processing for applications like smart cities and autonomous vehicles.



Thank You!!!